SLEEPING

Team from Macau PuiChing

YouTube

@SLEEPING_ROBOCUP

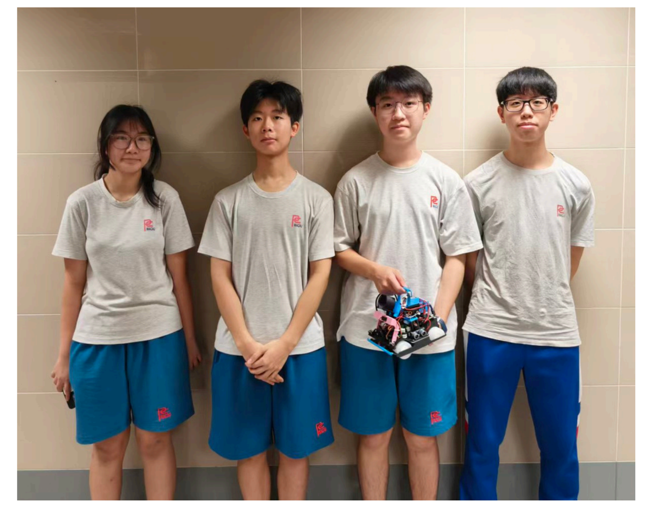
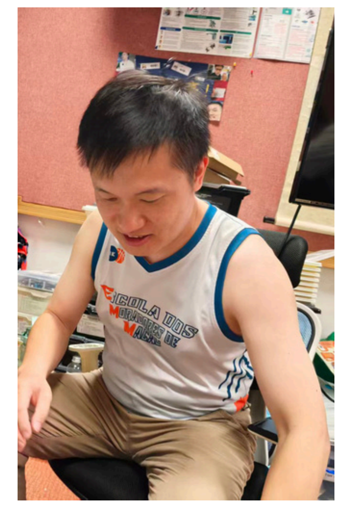
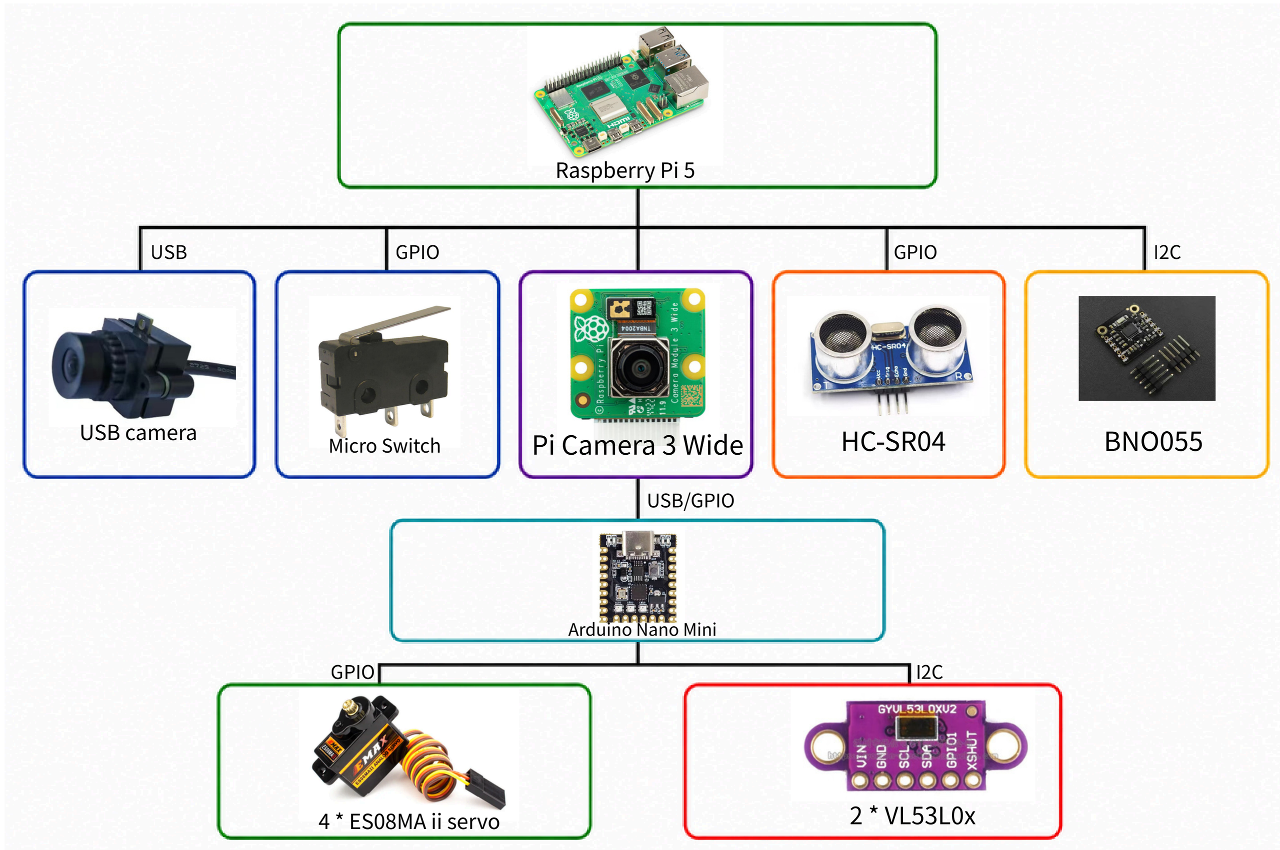


Photo of Team



Our Teacher

Hardware



HC-SR04

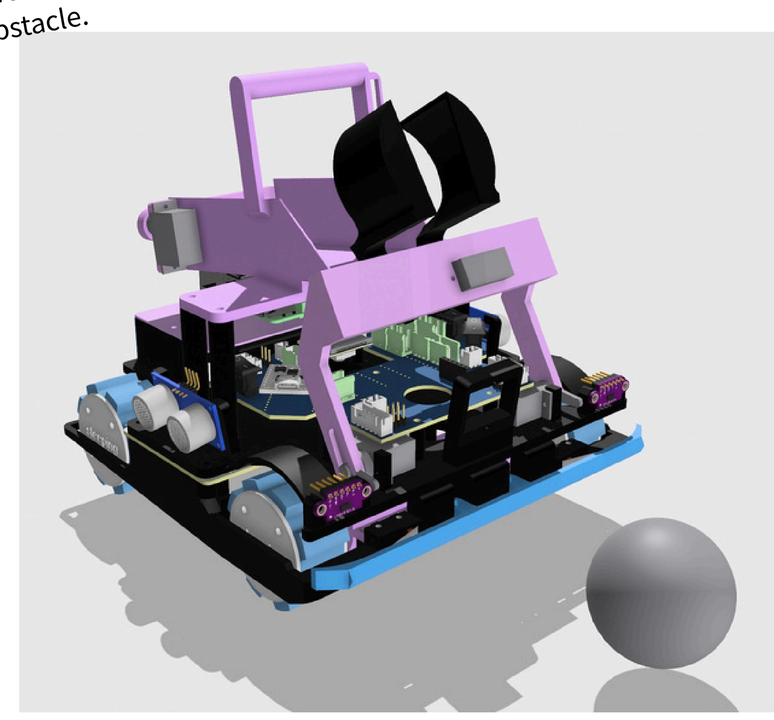
getting the distance between the robots and walls and obstacle.

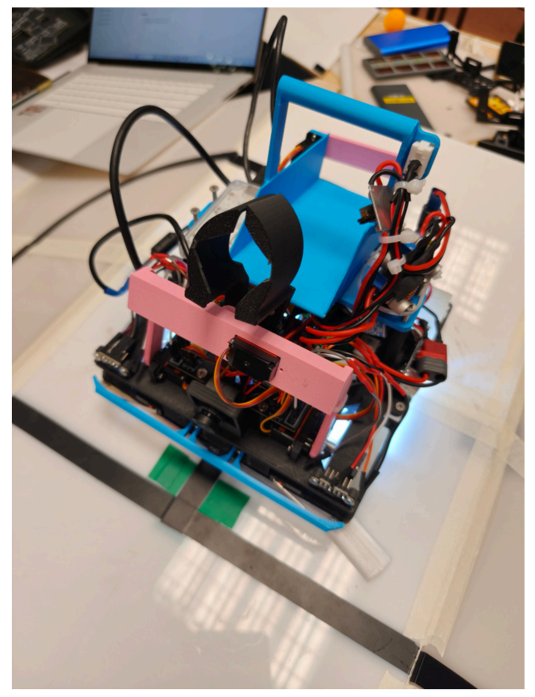
Raspberry Pi 5

Processes images from the USB and wide cameras to track the black line , detect green markers , and identify victim balls using a YOLO AI model.

USB camera

Detect victim in the evacuation zone.





4 * ES08MA ii servo

Control the arm and the picker to pick up the victims.

Micro Switch

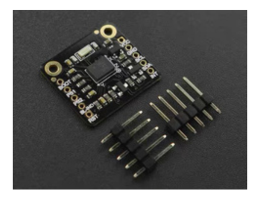
check the robot if it is touching the walls or obstacle.

2 * VL53L0x

getting the distance between the robots and walls.

Arduino Nano Mini

Control the arm and the picker to pick up the victims and reading data from vl53l0x.



BNO055

Obtain the robot's pose angles to detect inclines and declines road.

Software

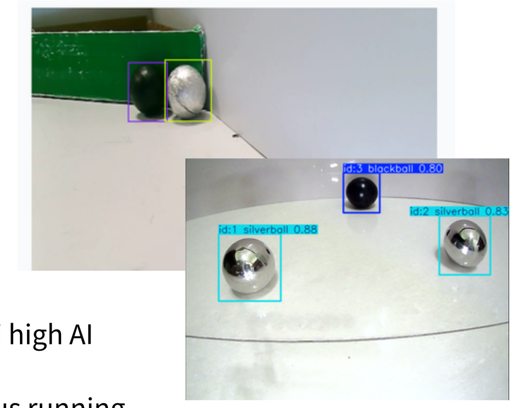
Python is a widely used, interpreted, and powerful high-level programming language. The code is highly readable and its syntax is close to English. Moreover, compared to other programming languages, Python commands are simpler to write and can achieve the same results with fewer codes.

ARDUINO

Arduino operates on bare-metal hardware without an operating system. This ensures instant I2C data fetching from the VL53L0X and immediate, jitter-free PWM signals to the servos for rapid obstacle avoidance.Generating precise servo pulses requires strict timing. Arduino handles this via hardware timers, ensuring smooth servo movements .

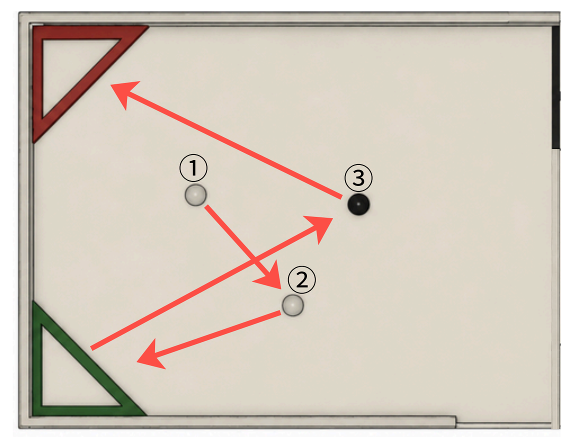
YOLO V8

We label more than 3000 image and train an Object detection model by using yolov8n to detect and collect victims in evacuation zone.



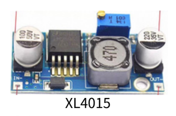
Google edge tpu give raspberry pi high AI performance.
4 TOPs of AI performance can let us running object detection over 20+fps

In the Evacuation Zone, our robot will pick up living victims(silver ball) first. Then put them in to the green evacuation point. After that, the robot will pick up dead victims(black ball) and put it in to the red evacuation point.



Power supply

Powering the raspberry pi 5



convert voltage to 12v for powering up the motors

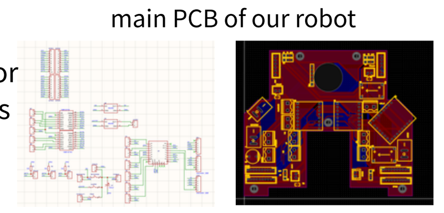
Giving power to arduino nano mini and LED light

convert voltage to 14v

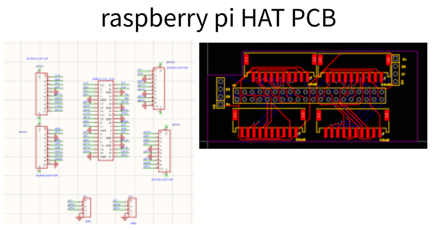
Powering supply of the whole robot

PCB

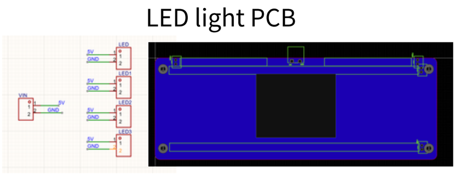
main PCB of our robot



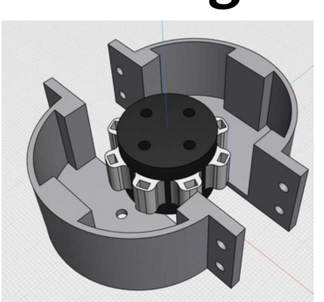
raspberry pi HAT PCB



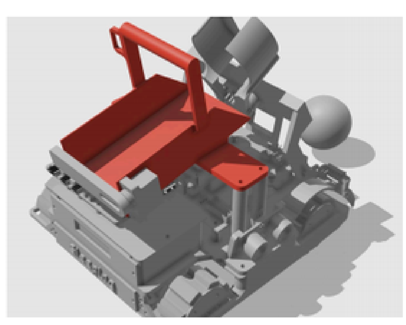
LED light PCB



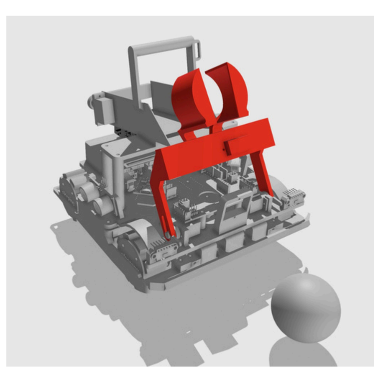
self making wheel



this wheel can help our robot pass ramps or speed bump easily



Vitctim Picker



We designed a arm for picking up the victims in evacuation zone.

Storage System

we designed a handle-equipped compartment designed to prevent balls from rolling out prematurely

Team members

Li Cheng Lok — Team Leader & Core Technical Architect Functions

as the overarching project coordinator and technical anchor. He plays a vital role across all dimensions of development, leading the mechanical design, hardware assembly, core Python programming, and systematic debugging to ensure seamless cross-system integration.

Lloyd — Main Software Programmer & Log Editor Acts

Acts as the intelligence developer for the robot's vision algorithms. He is responsible for selecting hardware components, drafting the engineering journal, and optimizing core algorithms—specifically the 24-slice line-following logic, HSV intersection marker detection, and the YOLO-driven evacuation zone behavior.

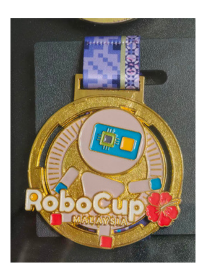
Mike.V — Documentation Specialist & Logistics Manager Serves

as the administrative backbone of the team. He functions to meticulously track milestone progress, manage task distribution calendars, oversee file version controls, and coordinate team logistics during intensive testing sessions at the innovation science center.

Wong Ioi Cheng — Lead CAD-Designer & Electronics Engineer Functions

as the primary mechanical and hardware designer. He utilized advanced 3D modeling to engineer the custom robot chassis, specialized claw mechanism, and ball storage compartment, while also leading the schematic layout and routing for the custom power-management and signal PCB boards.

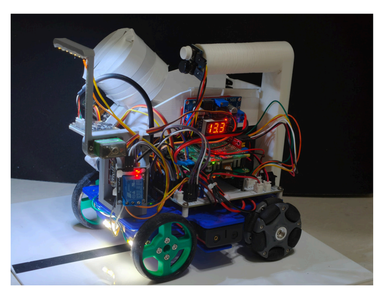
Award of us



🏅Malaysia Open line 2024

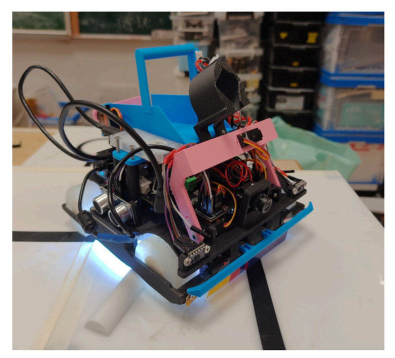
🏅China Open & HongKong & Singapore Open line 2026

Evolution of robot



RoboCupJunior Singapore Open 2025

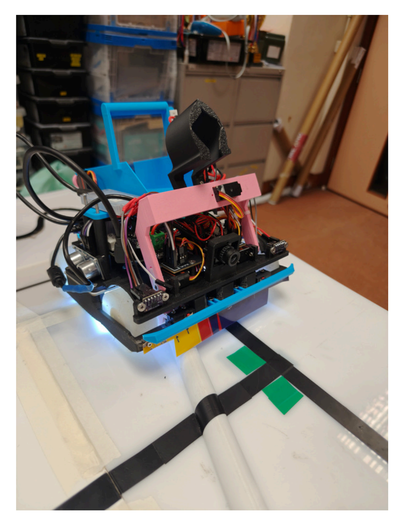
RoboCupJunior China Open 2026



RoboCupJunior World Championship 2026

Line Following

This part of the program divides the captured image into 24 horizontal slices and analyzes the position of the black line within each slice to determine its location.



To detect green intersections, the program follows a quick 4-step pipeline:
Color Filtering: It runs cv2.inRange using a targeted green threshold to create a binary mask.
Noise Cleanup: Consecutive erosion and dilation passes eliminate camera artifacts and smooth out the shapes.
Line Verification: It checks a small bounding box directly above the green contour; if the pixel average is dark, it confirms a real black line is present.
Direction Choice: Based on marker positions, it commands a Left, Right, or U-turn state.